

**THE SUSTAINABILITY OF NATIVE TREES IN THE GREENING
INITIATIVES OF THE PHILIPPINES**

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ABSTRACT

The nationwide tree planting of the Philippines through the National Greening Program (NGP) aims to conserve biodiversity with greater attention to the utilization of native species. As one of the major approaches to forest ecosystem restoration, the government including the academe and civil society organizations (CSOs) cooperate in promoting native trees as planting materials because they are naturally occurring species with a broader distribution range that are important in ecology, biodiversity, and economy. This review paper aims to 1) determine the parameters of sustainability of native tree species and the greening initiatives of the government, CSOs, and local communities; 2) analyze the strengths, weaknesses, opportunities, and threats (SWOT); and 3) develop a framework for the greening initiative sustainability of native trees. Printed and electronic materials from 2012 to the present were reviewed and analyzed to highlight the status of native trees as reforestation species with the NGP implementation since 2011. From 2011-2016, the total commodities planted comprised more than 14% native trees and 35% native or exotic timber and fuelwood species. The SWOT analysis revealed that native trees are prioritized as planting materials in reforestation but due to poor silvicultural practices and species-site mismatch, low survival rates were observed, and threats of natural phenomena limit their growth and survival. Partnerships for technical assistance and funding are opportunities to plant more native species that is independent of seedling availability and volume of stocks.

Keywords: native trees, biodiversity, forest ecosystem, restoration, quality planting materials

INTRODUCTION

Native trees are significant in the web of life (Tomas-Carig, 2020) specifically in reforestation and greening efforts to restore biodiversity and forest landscapes. They are referred to as naturally occurring species of trees. Indigenous tree species, on the other hand, are categorized as a type of native tree species being confined in a specific area or location in a country. As the Philippines records high species endemism with at least 19 genera of angiosperms comprising 53.2% of the total native species of the country (Pelser et al., 2011) with a total of 3,600 native plant species, the status of threatened plants is also increasing as presented by the updated inventory of threatened plant species and their categories (DENR Administrative Order 2017-11, 2017). Among the listed threatened species in the provision are the Philippine Mahogany group of Dipterocarpaceae including Narra (*Pterocarpus indicus* Willd.), thus, DENR further promotes the planting of native species. Native trees are the basis of our natural ecosystems (Lantican, 2015); provide food and habitat to wildlife better than introduced species; acclimatize well to their local surroundings making them more resilient than any other tree species (Tarriela, 2018); and possess the natural capacity to recuperate from destructive

pests and diseases and even harsh weather conditions (Lantican, 2015). They have a broader range of distribution compared to indigenous trees. Without direct or indirect human intervention, native plants live or grow naturally in a particular region while endemic species have restricted distribution within the geographic range (Lillo et al., 2019). Those species exclusively found in a particular geographic location, small or large are referred to as endemic plants or trees, also native species. Republic Act No. 9147 (2001) or the Wildlife Resources Conservation and Protection Act defines endemic species as “species or subspecies which is naturally occurring and found only within specific areas in the country.”

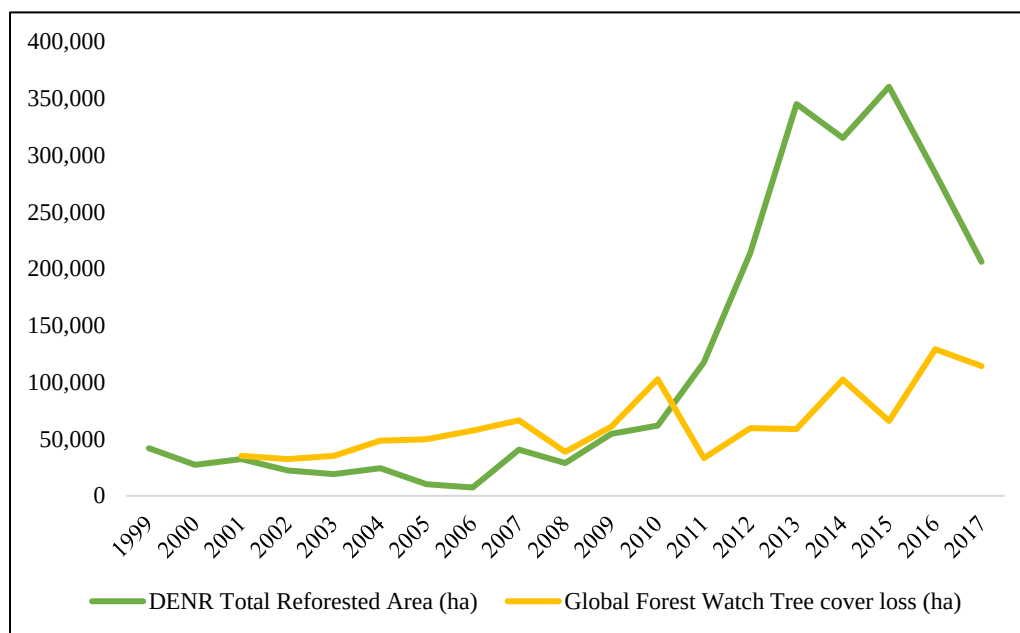
In 2020, the Food and Agriculture Organization (FAO) of the United Nations & United Nations Environment Program reported that plantation forests in the Asian region comprised more 60% of native species and the DENR expected an increase of 12% in forest cover relative to 2003 data of the country assuming 85% survival rate (Vista et al., 2016) of seedlings planted. While the DENR reported 1.67 billion various species planted in 2 million hectares nationwide with the NGP planted 25,848,659 seedlings in the 21,925 hectares planting site in 2019 (DENR-Forest Management Bureau,

2018), the non-government sector reforested 40, 883 hectares since the NGP launching (DENR-Forest Management Bureau, 2019). The Food and Agriculture Organization (FAO) in its 2015 Global Forest Resources Assessment, the Philippines ranked 5th of the largest forest area increase worldwide from 2010 to 2015 (Food and Agriculture Organization, 2016).

The NGP is the largest area reforested by various sectors as reported in the Philippine Forest Statistics by the DENR however, continuous loss of tree cover was also recorded by the Global Forest Watch during 2001-2017 (Figure 1). Hence, this review paper aimed to

discuss the status of greening initiatives in the Philippines in terms of the utilization of native tree species. Specifically, it 1) determined the parameters of sustainability of native tree species in the greening initiatives of the government, civil society organizations (CSOs), and local communities with their approaches to sustain the availability of native planting stocks; 2) analyzed the strengths, weaknesses, opportunities, and threats of native tree species; and 3) developed a framework for the greening initiative sustainability of native trees.

Figure 1. Trend of area reforested by the DENR (1999 to 2017) and tree cover loss in hectares as reported by the Global Forest Watch (2001 to 2017).



MATERIALS AND METHODS

An integrative approach based on Snyder (2019) was used in the literature review to collect, assess, and synthesize information about the utilization of native trees in the greening initiatives of the Philippine government. The inclusion criteria for identifying relevant literature were 1) the year of publication from 2012 to the present considering the NGP implementation which started in 2011; 2) the language of the articles or documents used which is English; and 3) the document type as peer-reviewed and open access for articles. The search strategy involved 4) typing keywords such as native trees or species, indigenous species, endemic species, reforestation, survival rates, and NGP in an authentication service (MyAthens research zone) that allowed access to subscription-based electronic resources, online databases, and other digital resources in various platforms, and in the Google search engine or via the Google scholar. The review of articles and documents was done 5) by initially reading the abstracts and making selections and then reading the full text ensuring that the inclusion criteria were met. The actual selection yielded thirteen journal articles, one article from conference proceedings, two review articles, six agency reports, four working papers, and four published books. To address the flaws

in limiting the search or data abstracted from the selected articles, 6) one approach was to visit the official websites of institutions and/or organizations involved in the NGP for online tracking of accomplishment reports, policies/guidelines, and published information about native trees and reforestation initiatives to support discussion and analyses of data gathered. Another way was 7) to include literature- earlier than 2012 as well as refer to the references of the selected articles to identify other articles that are relevant to the study. The information sourced from these materials were used for analyzing the extent of use of native tree species for reforestation in the Philippines and to develop a framework on the sustainability parameters of the greening initiatives.

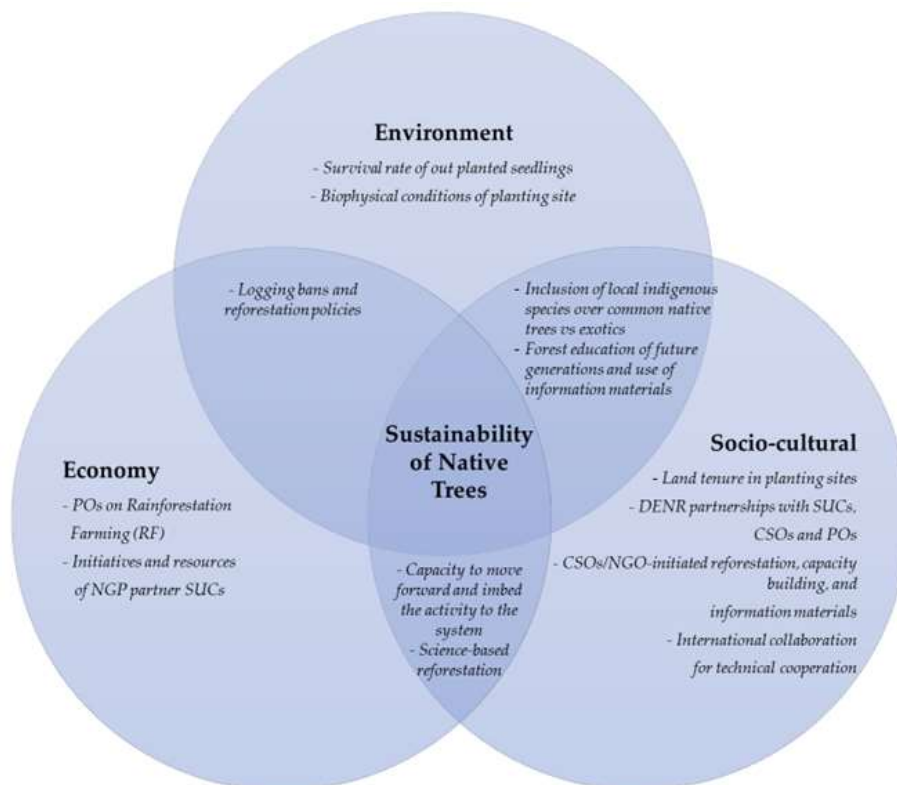
In general, the sustainability parameters of native trees were determined based on the pillars of sustainability: human, social, economic, and environmental. The various institutional support and organizational strategies for the nationwide greening program of the Philippine government were identified and grouped according to their contribution to the economy, environment, and socio-cultural aspects that are similar to the categories presented by the Food and Agriculture

Organization with the criteria and indicators for sustainable forest management of implementing national forest programs (Larrubia et al., 2017). The sustainability parameters were listed based on the basic requirement of measuring reforestation accomplishment which is survival rate then toward a conclusion which is moving forward that imbed the greening initiatives in the system.

RESULTS AND DISCUSSION

The major contribution of this paper is the assessment of thirteen (13) identified parameters of sustainability relative to the utilization of native tree species during the implementation of the national greening initiatives by the Philippine government with its partner NGOs, communities, and other government agencies. These are the factors or parameters to consider for the sustainability of native trees and the success of national greening initiatives in the country (Figure 2).

Figure 2. The factors that contribute to the sustainability of native trees in the greening initiatives of the Philippines.



1. Survival rate calculation of out planted seedlings

In assessing reforestation success, the survival rate of trees and the area planted versus the target area are the most common indicators used (Le et al., 2012), which is also one of the bases for reporting of the DENR including the number of seedlings planted in its nationwide greening accomplishment. The NGP set a standard of at least 85% survival rate for planted seedlings in NGP sites which was required for final payments under NGP contracts (Commission on Audit, 2019). But this is not the case for the survival rate among planted seedlings in various NGP sites. For instance, the impact study team formula of Balangue (2016) compared the DENR formula in estimating the survival rate of seedlings planted in selected NGP sites from 2011 to 2015 in Zambales, Dinagat Island, and Negros Occidental considering the seedling closing and opening stock, net surviving seedling, and mortality with a slight difference in the replanting values (annual vs cumulative per year). Based on their calculations, the mean survival rate of planted seedlings (either native or exotic) in the abovementioned sites was lower (58.18%) compared to the DENR estimates (86.50%). By dividing planted seedlings in the field by the sum of seedlings planted, the calculated

survival rate in Central Luzon, Cagayan Valley, and CARAGA region from 2011 to 2014 during NGP implementation was 69.35% based on DENR data in 2015 (Vista et al., 2016). Similarly, Luna (2016) reported a lower survival rate of 65.33% and 74.46% in Hinobaan, Negros Occidental and Zambales that included the native species Mangkono (*Xanthostemon verdugonianus* Naves ex Fern.-Vill.) planted in Dinagat Island in which there are no standards yet for its survival rate.

In the history of planting trees in the country by the Energy Development Corp. (EDC) under its BINHI Program in partnership with the DENR, it recorded a 40-60% survival rate of out planted native species while a 60-80% survival rate attained with locally indigenous species of trees specific to Negros Island Forest restoration program (Balangue, 2016). The assessment of Aguilos et al. (2020) proved that through domestication native trees can reach more than 80% survival rate for the first five years based on the 13 planted native trees (out of 33 species) in Lantawan Park in Kitcharao, Agusan del Norte (2014–2018). High survival rates and satisfactory growth were observed in most of the native species but the non-utilization of other native species in reforestation projects was due to problems in collecting planting

materials, most are rare, and lack phenological information (Aguilos et al., 2020). Domesticating commercially important native species could serve as a substitute to the large-scale growing of mixed and single species.

2. Biophysical condition of planting site

The NGP framework of planting sites targets the coastal, lowland, and upland ecosystems as areas for development that include forestlands, mangrove and protected areas, ancestral domains, civil and military reservations, urban areas under the greening plan of LGUs, inactive and abandoned mine sites (Calderon, 2014). The biophysical condition of these reforestation sites interacting with environmental factors such as temperature, light, pH, soil moisture, and topography, among others largely contributes to the growth, survival, and even mortality of planted seedlings.

The reforestation sites in Luzon, Sierra Madre, and Cordillera were assessed by Perez et al. (2020) using forest loss data derived from Landsat from 2010 to 2018 to compare the forest state before, during, and after the implementation of NGP. It was observed that many of the planted seedlings were in open fields where the plants have a lower chance of growth and survival. As for planting sites in reference to the source of

planting stocks, the NGP report of the Commission on Audit (2019) observed a low percentage of survival rate in some NGP sites due to the steepness of the terrain or unsuitability of the planting site and site-species mismatch during 2011 to 2015 implementation while the POs relate it to the distance of planting sites and natural calamities such as fire and typhoons. There are four climate types in the Philippines based on rainfall distribution and a great portion of the rainfall, humidity, and cloudiness are influenced by typhoons which have a great effect on the climate and weather conditions of the country (DOST-PAGASA, 2014). It is commonly practiced that tree planting in the country commences during the rainy months in June to December since warm temperatures trigger forest and grass fires during the summer season with human-induced fires such as the slash and burn practiced in the uplands.

The problem of finding appropriate native species for social forestry where public acceptability is vital and the planting endeavors that are often restricted to vacant farm lots and marginal lands reflects the negative relationship between environmental suitability indicators and community preference ranks (Chechina & Hamann, 2015).

3. NGP planting sites: tenured vs non-tenured

From 2013 to 2016, NGP planting sites were managed by the people's organizations (POs) who are holders of Community-Based Forest Management Agreement (CBFMA) or Protected Area Community-Based Resource Management Agreements (40%), POs without tenurial instruments (48%), and/or LGUs (Commission on Audit, 2019). However, the Commission on Audit (2019) revealed that the trend of low survival rates was attributed to non-tenured NGP sites such as those far areas that were abandoned after the maintenance and protection contract term. Based on the study of Martín (2007) in an upland agroecosystem in Leyte Island, the main factors that affect farmers' purpose to plant timber trees were the land tenure security and biophysical resources in terms of the land area and the number of lots managed by the household. Promoting the planting of slow-growing but high-value indigenous tree species like Molave (*Vitex parviflora* Juss.) and Lauan (*Dipterocarp sp.*) with high uptake rate forms the basis to review the arrangement of tenure duration in the country (Harrison et al., 2004).

Similarly, Milan (2020) recognized the significance of educating CBFMA / IFMA / SIFMA / CADT holders

employing Rainforestation Farming (RF) for forest restoration because it entails long investment, policy, and governance direction to ensure the tenurial security of protected areas, watersheds, mangroves, and important bird areas. Understanding such perspective reflects the responsibilities and collective ownership of the tenurial instrument.

4. Inclusion of local indigenous species over common native trees vs exotics

Species selection in the NGP depends on the planting objectives but preferably indigenous, native, and/or endemic species while, it should consider species-site-market matching for production zone and prioritizing indigenous, native, and/or endemic species for protection zones (Calderon, 2014). As of December 2021, there were 1,814,693,722 seedlings planted on 2,163, 271 hectares through the NGP (DENR, 2021). Of the total target in 2013-2016 (1,571,745 ha) NGP commodity roadmap of the DENR, only 14% was allocated to be planted with native species for protection forest (98,231 ha) and protected areas (127,407 ha) but in the 2013-2016 NGP accomplishment report, it failed to segregate which of the 1,154,541,951 seedlings are native species and exotics (Milan, 2020). The Commission on Audit (2019) data confirms that 14.14%

(approximately 235,000 hectares) of the NGP sites were planted with indigenous trees from CYs 2011-2016. The DENR- Forest Management Bureau produced at least 3.7 million indigenous tree planting materials in 2012 with 862,181 native tree species such as Agoho (*Casuarina equisetifolia* L.), Balitbitan (*Cynometra ramiflora* L.), and Kalumpit (*Terminalia catappa* Decne); and the DENR-Ecosystems and Research and Development Bureau reported 5.7 million native tree planting stocks produced with the Forest Foundation Philippines (FFP formerly PTFCF) and Foundation for the Philippine Environment (FPE) by 4.9 million, some 500,000 from Bukidnon Forest Incorporated (BFI), and 281,081 from 17 SUCs (Foundation for the Philippine Environment, 2013). Based on species category and germplasm type, there were 765 native seedlings produced in forest tree seedling nurseries located in Biliran, Leyte, Southern Leyte, and Samar (Region 8) and Bukidnon and Misamis Oriental (Region 10) in June and July 2013 (Gregorio et al., 2016).

Narra (*P. indicus*) was the top native species planted in most NGP sites of the ten most planted species nationwide including Mahogany (*Swietenia*

macrophylla King), Bakauan (*Rhizophora* sp.), Rubber (*Hevea brasiliensis* (Willd. Ex A.Juss.) Müll.Arg.), Falcata (*Paraserianthes falcataria* (L.) I.C.Nielsen), Coffee (*Coffea* sp.), Gmelina (*Gmelina arborea* Roxb. Ex Sm. In Rees), Cacao (*Theobroma cacao* L.), *Acacia mangium*, and Nangka or *Artocarpus heterophyllus* Lam. (Calderon, 2014). According to Milan (2020), Narra (*P. indicus*) and foreign or introduced species have been commonly used in reforestation activities in the country and the NGP has planted about 85 percent of fast-growing exotics while 11.9 percent (174,000 hectares) of the total 1.5 million hectares of the targeted area for NGP planted with native trees mixed with fruit-bearing seedlings mostly covered by partnership agreements between the DENR and SUCs as shared in a forum by the DENR former Undersecretary for Field Operation in 2016. Table 1 shows the common native trees produced by some SUC partners of DENR. Molave (*V. parviflora*) and White Lauan (*Shorea contorta* S. Vidal) were mostly propagated in their facilities and it could be noticed that Narra (*P. indicus*) was propagated asexually despite being a prolific seeder.

Table 1. Some priority species of SUCs propagated in clonal forestry nurseries.

SUCs	Priority Species for Clonal Propagation
Bicol University College of Agriculture (BUCAF), Guinobatan, Albay	Baligang, Kamagong (<i>Diospyros blancoi</i> A.DC.), Molave (<i>Vitex parviflora</i> A. Juss.)
Bohol Island State University (BISU), Tagbilaran, Bohol	Bayok-bayokan, Bitanghol (<i>Calophyllum blancoi</i> Planch. & Triana), Taguibokbok (<i>Gomphandra apoensis</i> (Elmer) Merr.), Molave (<i>Vitex parviflora</i> Juss.), White lauan (<i>Shorea contorta</i> S.Vidal), Palosapis (<i>Anisoptera thurifera</i> (Blanco) Blume), Manggachapui (<i>Hopea acuminata</i> Merr.), Gisok-gisok (<i>Hopea philippinensis</i> Dyer)
Mariano Marco State University (MMSU), Ilocos Norte	Molave (<i>Vitex parviflora</i> A. Juss.), Dao (<i>Dracontomelon dao</i> (Blanco) Merr. & Rolfe), Supa (<i>Sindora supa</i> Merr.), Lanete (<i>Wrightia pubescens</i> R.Br.), Ipil (<i>Intsia bijuga</i> (Colebr.)

	Kuntze), Kalumpit (<i>Terminalia macrocarpa</i> Decne.), Narra (<i>Pterocarpus indicus</i> Willd.), Kamagong (<i>Diospyros blancoi</i> A.DC.), Yakal (<i>Shorea astylosa</i> Foxw.), Saplungan (<i>Hopea plagata</i> (Blanco) S.Vidal), Guijo (<i>Shorea guiso</i> (Blanco) Blume), Palosapis (<i>Anisoptera thurifera</i> (Blanco) Blume), White lauan (<i>Shorea contorta</i> S.Vidal)
Ramon Magsaysay Technological University (RMTU), Iba, Zambales	Batino (<i>Alstonia macrophylla</i> Wall. ex G. Don), Yakal (<i>Shorea astylosa</i> Foxw.), Kalingag (<i>Cinnamomum mercadoi</i> S.Vidal)
Southern Luzon State University (SLSU), Lucban, Quezon	Makaasim (<i>Syzygium nitidum</i> Benth.), White lauan (<i>Shorea contorta</i> S.Vidal), Toog (<i>Bischofia javanica</i> Blume), Kalumpit (<i>Terminalia macrocarpa</i> Decne.), Bagtikan, Molave (<i>Vitex parviflora</i> A. Juss.), Batikuling (<i>Litsea leytensis</i> Merr.) , Igem (<i>Dacrycarpus</i>

	<i>imbricatus</i> (Blume) de Laub.), Kolalabang <i>(Saurauia latibractea</i> Choisy in Zoll.), Dita <i>(Alstonia scholaris</i> (L.) R.Br.), Almaciga <i>(Agathis alba</i> (Rumph. ex Valmont) Foxw.), Aunasin, Malabayabas (<i>Tristaniaopsis</i> <i>decorticata</i> (Merr.) Peter G. Wilson & J.T. Waterh.) , Malabulak (<i>Bombax ceiba</i> L.), Bignay (<i>Antidesma bunius</i> (L.) Spreng)
Tarlac Agricultural University (TAU), Camiling, Tarlac	Yakal (<i>Shorea astylosa</i> Foxw.), Lanete (<i>Wrightia arborea</i> (Dennst.) Mabb.)
Visayas State University (VSU), Baybay City, Leyte	Yakal (<i>Shorea astylosa</i> Foxw.), Molave (<i>Vitex</i> <i>parviflora</i> A. Juss.), Narra (<i>Pterocarpus indicus</i> Willd.), Bagtikan (<i>Parashorea malaanonan</i> (Blanco) Merr.), Mangkono (<i>Xanthostemon</i> <i>verdugonianus</i> Náves ex Fern.-Vill. in Blanco), White lauan (<i>Shorea contorta</i> S.Vidal), Red lauan (<i>Shorea negrosensis</i>

	Foxw.), Dao (<i>Dracontomelon dao</i> (Blanco) Merr. & Rolfe), Bitanghol (<i>Calophyllum</i> <i>blancoi</i> Planch. & Triana)
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Promoting the use of indigenous species in the NGP is important to the global recognition in ecosystem restoration, thus it is relevant to plant native species (Gregorio et al., 2016). Reforestation programs may have improved biological diversity by the planting of native tree species per location as well as neutral to positive impacts on soil and water quality like in extreme flood events and landslide occurrence (Vista et al., 2016). Field observations on the growth of *S. contorta*, *P. indicus*, and *V. parviflora* proved that they grow at least as quickly as some exotic species with different performance in terms of productivity and is similar or even better than the exotics because they have a considerable extent of intercropping benefit compared to single cropping system and *V. parviflora* was the most promising agroforestry native species at intermediate densities (Martín, 2007). Also, the use of dominant species as planting materials in disturbed riparian areas can improve success rates in reforestation since these species have the best survival strategy

in a such ecosystem that may be applicable to other environment types (Pasion et al., 2020). The abundance of native tree species in these sites has comparable effects to the biophysical conditions of the environment, diversity and even ecosystem functions such as soil nutrient cycling and pollination.

5. DENR partnerships

The NGP strategies of the DENR involved the 1) harmonization of initiatives with partner NGOs – FPE and Forest Foundation Philippines (FFP formerly PTFCF), 2) provision of incentives to people's organizations, and 3) maximization of available S & T through clonal nursery, biofertilizers and pesticides, and GIS mapping and geotagging (DENR, 2021). Thus, the DENR entered into partnership agreements with GAs, NGOs, and POs. Pursuant to Executive Order No. 26, s. 2011, the average cost of NGP implementation (2011 to 2016) is PhP21,421 per hectare: PhP16,421.00 for the establishment activities,

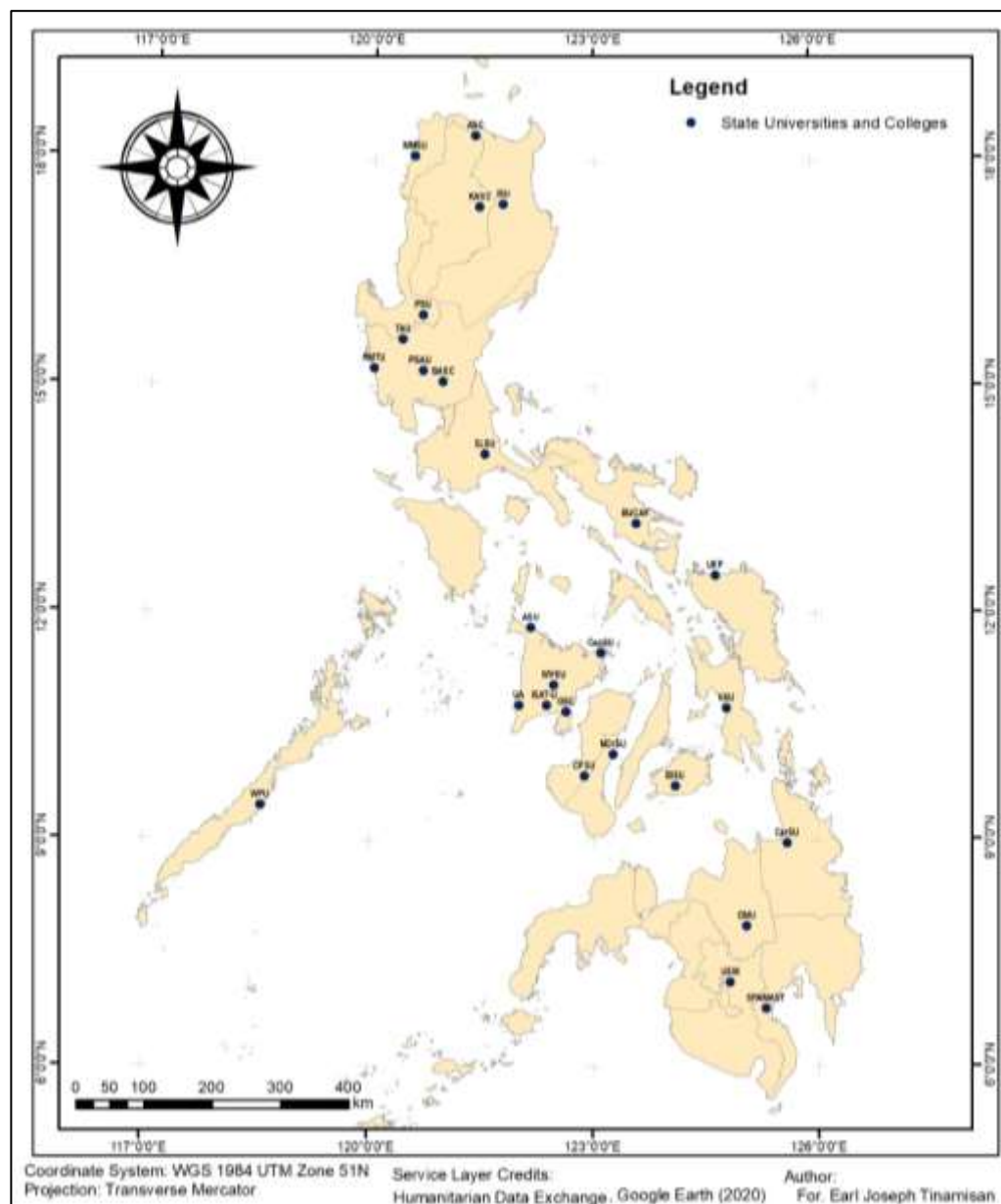
PhP6,000.00 for maintenance and protection costs distributed in three years: PhP1,000 for Year 1, PhP3,000 for Year 2, and PhP2,000 for Year 3 (Vista et al., 2016). The establishment activities involved the production of seedlings; survey, mapping, and planning; planting site preparation; awareness campaigns; conveyance; and mobilization.

In 2012, the PTFCE/FFP and FPE forged a partnership agreement with the DENR where CSOs commitment to NGP was 45,608,000 seedlings for upland areas and they were able to cover 25,271 hectares of land with native tree species (Milan, 2020). FFP and FPE are member NGOs of the Rain Forest Restoration Initiative (RFRI). Meanwhile, 651 POs were engaged in the IEC, social preparation, joint planning and site identification; seedling production, site preparation, planting, and maintenance and protection (including replanting) activities in the NGP aimed at harvesting and utilization in the future (Calderon, 2014). To augment the NGP requirements of quality planting materials of native tree species, the clonal propagation technology using vegetative plant parts of native tree species was recommended by the DENR-Ecosystems Research and Development Bureau (ERDB) with the application of mycorrhiza technology

developed by UPLB-BIOTECH (MykoVAM) and DENR-ERDB (Hi-Q VAM 1). The DENR established 22 regional clonal nurseries nationwide supported by the ERDB regional research centers such as the Manleluag Spring Protected Landscape (MSPL) Clonal Nursery of DENR Region I in Brgy. Malabobo, Mangatarem, Pangasinan. Twenty-seven (27) agriculture and forestry SUCs were granted by the DENR P3.5 million each for the establishment of clonal facilities and for the initial production of 60,000 native seedlings from 2012-2014 and 20,000 clones of priority forest tree species yearly from 2015 onwards (ERDB-DENR, 2017) forged through a memorandum of agreement (Figure 2). The partner SUCs have contributed about 12% of the total 1.5 million ha of NGP-targeted area by the DENR (Milan, 2020). The clonal propagation technology is highly suitable for producing planting stocks that are independent of seed, seedling volume, and seasonal availability of the target species for tree plantation (Gutierrez et al., 2020). The easy method of propagation and short rotation period allows the establishment of forest plantation with greater success, however, plantation failure, diversity loss at the forest and landscape levels, and genetic gain risk are perceived from using clones in forestry (Wu, 2019).

The above strategies of the DENR may have had direct and/or indirect influence on the promotion and sustainability of using native trees in NGP. However, the crucial factors in the success of the recipients of these contracts are 1) their preparedness for program implementation and 2) the convergence of these agencies with the private sector (Commission on Audit, 2019).

Figure 3. Clonal forestry nurseries of SUCs operating nationwide.



6. Initiatives of Non-Government Organizations

Non-government organizations in the Philippines play a very active role in conserving the 3,600 native tree species by promoting and propagating them in various tree-growing activities in selected sites across the country. Initially, FPE, FFP and the BFI were the NGOs engaged in NGP. The partnership agreement of DENR with FPE and PTFCF produced an estimate of more than 10 million seedlings that promoted the country's native tree species (Milan, 2020). For the past 30 years, the Philippines was one of the highest country-level richness of species commonly planted by tree planting organization projects with 97 species such as *Tectona grandis* L., *A. heterophyllum* Lam., *Samanea saman*, *Anacardium occidentale* L., *Tamarindus indica*, *Annona muricata* L., and *Syzygium cumini* (L.) Skeels (Martin et al., 2021).

The Philippine Native Plants Conservation Society, Inc. (PNPCSI) and the Philippine Native Tree Enthusiasts (PNTE) are two of the active non-profit environmental organizations in the promotion of awareness and conservation of native tree species and their habitats. PNCSI is duly registered under the Securities and Exchange Commission and Bureau of Internal Revenue in 2008 whose founding

president is the late Leonard L. Co. It has about 120 members from the academe, non-governmental organizations, government agencies, horticulture sector, garden clubs and plant hobbyists and enthusiasts (The Philippine Native Plants Conservation Society Inc. and Southern Luzon State University, 2017). The organization was able to list down 391 native tree nurseries nationwide being supervised by the FFP, Haribon Foundation, Inc., and FPE (Table 2). It hosts several national and international symposia nationwide every other year since November 2011 to commemorate the first death anniversary of its founder. The first symposium was held at the National Museum that served as the platform for the launching of "Co's Digital Flora of the Philippines," it is the most comprehensive and authoritative website on Philippine plants based on Mr. Co's updates of E. D. Merrill's (1923) checklist; and the RFRI's 1st Leonardo L. Co Award for Best Native Forest Tree Nursery (PhilStar Global, 2011). For PNTE, the social media has been a useful tool to discuss and share knowledge on Philippine native trees such as tree identification, seed germination, tree growing techniques, and problems encountered leading to mainstreaming of native trees. PNTE is an informal group in Facebook which was formed in August 2013, an environmental-civic organization

separate from PNPCSI with about local and abroad with a common 20,000 members from diverse passion and love of native trees. backgrounds and sectors of society both

Table 2. Nurseries of native tree species based on PNPCSI record in 2014.

Regions	PTFCF/FFP		FPE	Haribon	PNTE	Total
	Regular Projects	NGP Partners				
CAR Cordillera Administrative Region	3	7	7			17
NCR National Capital Region					3	3
I Ilocos Region	2			7		9
II Cagayan Valley	7		24		1	32
III Central Luzon	1	1	4	1		7
IV-A CALABARZON	2		32	4	1	39
IV-B MIMAROPA	4	4	43	1	1	53
V Bicol Region	2	1	18		1	22
VI Western Visayas	5		11			16
VII Central Visayas	4		9			13
VIII Eastern Visayas	2	3	20			25
IX Zamboanga Peninsula	2		41		1	44
X Northern Mindanao	2		49			51
XI Davao Region	1		11			12
XII SOCCSKSARGEN	2	4	30			36
XIII CARAGA Region	1		11			12

7. Science-based reforestation

The BINHI Tree for the Future Program or simply BINHI is a reforestation approach by the Energy Development Corporation (EDC). This science-based reforestation initiative aims to save and protect ninety- six (96) of the most threatened native tree species known in the country that are nurtured with the help of one hundred ninety-one (191) partner institutions including students, employees, executives, and local communities from 16 regions across the country (Malabrigo, Jr., 2016). A public-private partnership was also forged by the DENR with the EDC through its BINHI. The nationwide greening program of EDC has planted 6.4 million seedlings in 9,449 hectares of denuded forests, restored inside geothermal reservations, and other watershed areas (Energy Development Corporation, 2018). Seedlings from BINHI had a 98% survival rate (Malabrigo, Jr., 2016) and among the rescued native trees are Narra (*P. indicus*), the Dipterocarps, and many premium species. EDC published a book, "BINHI Tree for the Future" and launched a website to track and document the status of planted native trees all over the country (tree locator) to raise awareness among various stakeholders.

Like BINHI, the Buhay Punlaan of Haribon Foundation serves as a native

tree nursery, laboratory and training facility that constantly communicates the importance of biodiversity conservation in line with the foundation's advocacy (Haribon Foundation, 2020). It is an outdoor classroom for learning how to bring back forests using native tree species that is located at the Caliraya-Lumot Watershed in Lumban, Laguna. To date, more than 73,140 native tree seedlings have been produced since 2009, with the help of about 4,000 volunteers and planted 15,685 native trees in a 3-hectare site (Haribon Foundation, 2020).

There has been an increasing interest among Filipino researchers on studying the probabilities of occurrences of native tree species in terms of current and future climate combined with other environmental variables for suitable habitat range. The habitat suitability maps and species distribution models generated from these studies could provide practitioners to formulate appropriate science-based adaptation policies, strategies, and measures as well as early warning estimates of how climate change may affect the distribution of threatened native forest tree species that could enhance their resilience and natural ecosystems to current and future climate (Garcia et al., 2013; Paquit et al., 2017; Torres et al., 2016; Tumaneng et al., 2019).

8. People's organizations and local community efforts

One way to conserve threatened Philippine native species and sustain its utilization as planting materials is the nationwide adoption of Rainforestation Farming (RF). In the early 1990s, RF was developed as an agroforestry system by the Visayas State University (VSU) and the German Agency for Technical Cooperation using native trees in rehabilitating degraded landscapes and restoring key ecosystem services and functions that provides forest-dependent communities with an alternative source of livelihood (Rain Forest Restoration Initiative, 2022). It was later adopted by the DENR through Memorandum Circular 2004-06 as an official reforestation strategy before the launching of the NGP. The RFRI was given initial funds by the FFP and FPE for mobilization and capacity building of RF adopters across regions while the Environmental Leadership & Training Initiative (ELTI) sponsored RF training to various groups in which it could be adapted to specific contexts (Milan, 2020). The approach in RF involves the planting of pioneer tree species in the initial phase and then followed by late successional species in the succeeding years which imitate the natural succession patterns and create a diverse, stratified forest (Environmental Leadership and Training Initiative,

2012). With the aim to further promote native species for forest restoration in the Philippines, it created an information portal of RF sites and nurseries, and other resources over the internet.

Under the NGP, demo farms on RF in Pilar, Camotes in Cebu Province planted different species of native trees such as Narra (*P. indicus*), Magtalisay (*T. catappa*), Molave (*V. parviflora*), White lauan (*S. contorta*), Kalumpit (*Terminalia macrocarpa* Decne.), Tugas, and Kamagong (*Diospyros blancoi* A.DC.) in combination with fruit trees like jackfruit, Santol (*Sandoricum koetjape* (Burm.f.) Merr.), tamarind, guava, and exotic tree species like Mahogany. Individual adopters of RF in Leyte, Cebu, and Negros Oriental planted the same species as mentioned earlier with Dipterocarps such as Gisok-gisok (*Hopea philippinensis* Dyer) and Yakal (*Shorea astylosa* Foxw.); Toog (*Bischofia javanica* Blume), Dao (*Dracontomelon dao* (Blanco) Merr. & Rolfe), Antipolo (*Artocarpus blancoi* (Elmer) Merr.), and Bitanghol or *Calophyllum blancoi* Planch. & Triana (Fernandez & Bande, 2019).

The utilization of native tree species was assessed by Chechina & Hamann (2015) in the North Negros Natural Park and found that Dipterocarpaceae, Fabaceae and Lauraceae with five or

more highly valued species each were ranked highest in social importance. Local communities preferred native trees like the Dipterocarps known as premium hardwood species for construction and furniture purposes, Fabaceae species as soil nitrogen-fixers, and Lauraceae species for their timber. Late-successional native tree species were highly valued and were commonly used in local reforestation programs (Chechina & Hamann, 2015). However, the results of the study showed a negative correlation between inferred ecological suitability and social preference ranks of species suitable for open-field reforestation that have commercial values.

9. Policy initiatives

Since the 1990s, the national logging bans and the reforestation policies have contributed to the Philippine forest transition particularly in increasing areas of planted forests in total forest cover while the primary forests did not change much in area extent (Pichler et al., 2021). The DENR has promoted Rainforestation Farming (RF) in restoring the original vegetation stand that promotes and conserves the biological diversity in the area by facilitating the natural process of succession using only indigenous and endemic tree species as planting materials (DENR Memorandum

Circular No.2004-06 Guidelines in the Integration of Rainforestation Farming Strategy in the Development of Open and Denuded Areas within Protected Areas and Other Appropriate Forest Lands, 2004). Relative to the use of native tree species as planting stocks, a network of CSOs wrote a letter to the DENR in 2015 that emphasized their strong position that government funds should not be used to procure and maintain exotic species for any reforestation programs and objected the targeted 10 percent native species of the implementing guidelines in MC 2011-01 or EO 26 (Milan, 2020). With this, the DENR implemented “a phased approach to a shift from use of exotic to indigenous species, starting with establishment of seed production areas and clonal nurseries especially of species for the production areas and engaged the PTFCF's and FPE's assisted CSOs in the production and planting of indigenous forest species pursuant to the MOA among DENR and these NGOs,” (DENR Memorandum Circular No.2012-01 Implementation of The National Greening Program, 2012). At the local level, an ordinance in Quezon City mandated all its barangays to include the planting of native trees in their respective environmental programs and projects in their areas of jurisdiction in support of NGP (City Council Ordinance No. SP - 2359, s.2014 The Native Re-Greening Ordinance of

Quezon City, 2014). It is to promote ecological balance, a wholesome environment and other purposes that further encourage citizenry involvement.

In 2015, the NGP expansion was signed consistent with the updated Master Plan for Forestry Development (2016 – 2028) “in order to harmonize all forest development activities that will encourage and enhance the development of forest plantations including forest parks, with greater participation from the private sector, local government units and organized upland communities” (Executive Order No. 193, 2015).

10. International collaboration for technical cooperation

Rainforestation Farming (RF) as a form of community forestry with native tree species through the Visayas State University in collaboration with the German Technological Cooperation has shown success in the Philippines (Chechina & Hamann, 2015).

The ELTI is a joint initiative of Yale School of Forestry and Environmental Studies and Smithsonian Tropical Research Institute co-sponsored conferences/workshops on NGP's utilization of native trees with FPE, PTFCF/FFP, RFRI and DENR in 2014

and 2015. During the NGP summit in 2014, ELTI put up posters as an approach to restoring the forests of the Philippines for communities, biodiversity and ecosystem services, and provided additional information about their work in scaling up Rainforestation (Environmental Leadership & Training Initiative, 2014). With RFRI and the Institute of Biology, University of the Philippines (UP) Diliman, it co-sponsored the conference on, “Mainstreaming Native Species-based Forest Restoration” held on July 15-16 at UP-Diliman that presented the success stories of forest restoration using the Miyawaki method, Rainforestation, Upland AF system, among others (Neidel et al., 2012).

11. Capacitating young minds and future generations

In Pilar, Camotes, Cebu where the DepEd primary and secondary schools and the island of Pilar, Camotes in general were adopters of NGP that planted Narra (*P. indicus*), Magtalisay, and Molave (*V. parviflora*) with fruit trees like jackfruit and tamarind in their school in a forest. White Lauan (*S. contorta*), Narra (*P. indicus*), Magtalisay, Kalumpit (*T. microcarpa*), Tugas, Kamagong (*D. blancoi*), and fruit-bearing trees like jackfruit, Santol (*S. koetjape*), and guava, among others in demo farms on rainforestation and

biodiversity restoration (Labastida-Martinez & Fernandez, 2020). The results and findings of Abella & Cutamora (2019) on the implementation of NGP in a secondary school in Simala, Cebu for 2016-2017 has been effective based on the evaluation of those who participated in the programs, project operations, impacts, and social effect (using the ABCD model) which correspond with the realities of minor differences only.

The Green Convergence for Safe Food, Healthy Environment and Sustainable Economy, and Hortica Filipina Foundation, Inc. published the Philippine Native Trees book 101 in 2012, book 202 in 2015, and book 303 in 2018 with support from the BINHI Project of the Energy Development Corporation, Lopez Group Foundation, Inc., and Forest Foundation Philippines. The books put together essays of non-professional writers (with so much passion for native trees) for non-botanists. The books provide information on the familiar name of the species, ecology class, height, trunk, tree type, conservation status, growth, light requirements, leaves, flowers, fruit, propagation, use, and other unique characteristics or descriptions. The Book 101 contains 108 personal stories of our native trees by 104 writers that begin with Agoho (*Casuarina equisetifolia* L.) and up to Yakal

Saplungan (*Hopea plagata* (Blanco) S.Vidal). It showcases photographs of native trees that were mostly taken in metropolitan areas to encourage their continued use in cities and urban centers since not all people visit the wild to appreciate them. Book 202 contains 123 native tree stories with a glimpse of the life of the late Leonard L. Co, a prominent Filipino botanist and plant taxonomist. The book also highlights the success stories of groups that advocate growing native trees within their vicinities. It contains essays for Agoho Palawan (*Ceuthostoma palawanense* L.A.S.Johnson) up to Zambales pitogo (*Cycas zambalensis* Madulid & Agoo). The third book contains 145 native tree species with its scientific name and derivation (Fernando, et al., 2018). It paid tribute to the memories of known advocates of native trees such as Dr. Antonio C. Manila, Dr. Roberto E. Coronel, and Dr. Benito S. Vergara. An interesting content of Book 303 is the feature of agarwood or *Aquilaria* species by Oliver Conda as these trees are on the verge of extinction due to agarwood oil trade. To date, the Green Convergence partnered with First Philippine Holdings Corporation in launching five story books on native trees for Filipino children. The stories were inspired by the essays written in Books 101 and 202 on Bignay (*Antidesma bunius* L.), Dapdap (*Erythrina variegata* L. in

Stickm.), Dita (*Alstonia scholaris* (L.) R.Br.), Ilang-ilang (*Cananga odorata* (Lam.) Hook.f. & Thomson), and Patalsik-pula (*Decaspermum blancoi* S. Vidal). In addition, a guidebook on developing and managing forest plantations for classroom instruction and field operations was published by the DENR-ERDB in 2010 showcasing the different techniques or protocols used in producing quality planting stocks of native forest trees like the application of clonal forestry and mycorrhiza technology. It also contains information of potential seed production areas per region of the country, and nursery management techniques for native forest tree species, among others that will assist tree growers and forest plantation managers nationwide.

12. Own initiatives and resources of NGP partner SUC on clonal forestry

SUCs have been an integral part of the NGP convergence initiative for the production and delivery of seedlings through hedge garden of native forest tree species and clonal facilities for vegetative propagation. For instance, the Southern Luzon State University (SLSU) has engaged in several Memorandum of Agreements with the DENR from 2012 to 2017 of more than P10 million worth of quality planting materials or 859, 290 seedlings

delivered to different municipalities of Quezon province and other NGP sites in Bataan, Nueva Ecija, and Pangasinan. However, funding from DENR is not enough for SUCs like the SLSU to enable and sustain the clonal facility to operate on regular basis plus employment of nursery staff and research assistants. SLSU's own initiative was to spend about a million pesos over and above the P3.5 million grant from DENR to complete the clonal forestry complex construction, maintenance, and operation (Cruz, 2013). Additional resources were initiated by the university through the provision of a regular faculty to supervise the project with an annual budget for seedling production (including personnel) since 2018; a research fund from 2016 to 2017 and 2018-2019; and an extension grant from the university in partnership with the municipal government of Paete, Laguna from 2018 to 2019. At present, the facility continuously produces native forest tree seedlings as an income-generating project that provides quality planting materials to prospective recipients and clientele with ongoing research and extension activities on native tree species in Quezon Province.

13. Capacity to move forward and imbed the activity into the system

Beyond the production of seedlings to augment planting materials supply, the SUCs clonal facilities catered to the needs of other stakeholders in terms of instruction, research, and extension. The SLSU Clonal Forestry facilities are being utilized for class instruction, research projects, and extension education of faculty and students at the university (Cruz, 2013). SLSU constructed an information bulletin of research conducted, and extension education thru training and seminars to target beneficiaries. It has developed a scientific way of producing high-quality planting materials in a short period of time making them available during the planting season. In research and development of seedling production, 50 forest tree species indigenous in Mt. Banahaw de Lucban were produced. In fact, the propagation using stem cuttings of Batikuling (*Litsea Leytensis* Merr.) is now being practiced by the stakeholders of Paete, Laguna, the Carving Capital of the Philippines (Gutierrez et al., 2020). Wood of *L. leytensis* is the raw material for carving statues and architectural embellishments that are known nationwide and worldwide. Unfortunately, it is now a threatened Philippine tree species since the wood supply is not abundant and over-

exploitation may have decreased it to a critical level. In 2020, SLSU was tapped to present its successful utilization of clonal propagation technology at the 5th National Research and Development Conference in November 2020 led by the Department of Science and Technology in collaboration CHED, DA, DENR, DICT and DTI.

The clonal facility of SLSU conducts research and development initiatives (including quality planting material production) with funding support of the government that brought about change and/or significant impact on society and the environment. It promotes the conservation of forest tree species through knowledge generation, protocol development, and extension on the production of planting stocks that considers the local communities as stewards of our vulnerable ecosystem.

SWOT Analysis for Native Trees in Greening Initiatives

The future and sustainability of native tree species for reforestation could be analyzed on the strengths, weaknesses, opportunities, and threats (SWOT) of approaches by the government, local communities, and civil society organizations (Table 3). Native tree species should be considered as priority planting materials in any tree planting or tree growing activities because they

naturally exist in our surroundings. The establishment of various species and plantations of native species offer multiple ecosystem goods and services simultaneously rather than traditional monoculture systems in a large-scale (Le et al., 2014). In fact, studies have shown their comparative advantage in forest regeneration, biodiversity conservation, and ecosystem management. For instance, a relatively high density of native species in forested landscapes is important to protect site-scale conservation measures, such as the designation of protected areas as a possible source of endemic and threatened tree species e.g., Dipterocarpaceae species for reforestation programs (Pasion et al., 2020). The management of protected areas is important to enhance biodiversity and protect native species from destructive human use.

Strengths

During the initial implementation of the NGP, it prescribed the use of tree species naturally growing (native) species in the country or local area as planting materials to ensure higher survival rates following: 1) appropriateness to predominant site conditions, 2) planting objectives, 3) availability of planting stocks, and 4) commercial value for market (DENR et al., 2012). Also, the Philippine Master

Plan for Climate Resilient Forestry Development 2016-2028 strategies to make the country responsive to climate change challenges, it recommended the use of resilient native species in the reforestation of critical watersheds, protected areas, and agroforestry sites; and planting of native species that are fast-growing in production forests (DENR, 2016). The Implementation Manual for NGP introduced 34 indigenous species from 19 families as planting materials (DENR et al., 2012) alongside the DENR-ERDB established 75 seed sources through the Seed production Areas and individual plus trees to produce quality planting materials in support of the NGP and its expansion based on section 8.7 of the DENR Memorandum Circular No. 2011-01 (DENR-ERDB, 2019).

As the NGP emphasized the priority for some native trees complemented with fruit trees, many CBFM farmers preferred it because some native tree species produce timber and several multiple uses (Ota et al., 2020) as these have been utilized for generations as a source of food, fiber, fuel, and folkloric medicine by locals. Native tree timber species are not invasive, adapt well to the environment, are mostly accepted by local people, are widely used in existing applications as supported by current local knowledge, and are relatively significant in the local culture

as based on a simple evaluation of local species in terms of yield for smallholder farms in the Philippines (Santos Martin & van Noordwijk, 2011).

The adopters of school environmental projects of DepEd revealed that it is the absolute commitment and passion for the environment that boost their willpower (Labastida-Martinez & Fernandez, 2020). Throughout the years, the SUCs clonal facilities have had the capacity to move forward and imbed the clonal propagation technology in the R & D and extension education system of the academe while the NGOs' creativity to promote native species through IEC materials, seminars/conferences, and capacity development activities attracted funding from local support groups and international agencies.

Weaknesses

The NGP budgeted 28% of the total cost for maintenance and protection is smaller than the prevailing budget of PICOP Resources (PhP40,000 per hectare), ABS-CBN Bantay Kalikasan Foundation (PhP 78,000 per hectare) and UP Land Grant (PhP75,000 per hectare) according to Vista et al. (2016). In line with SUCs clonal facilities, the funding to sustain these activities is a concern to enable the clonal facility to operate after MOA with DENR is

terminated coupled with the low marketability of produced clones and the high cost of maintenance. For the DepEd teachers implementing NGP activities at school and in their community, the challenges encountered show that many residents and students still showed pessimistic attitudes towards environmental initiatives which is the symptom of a bigger problem that is rooted in poverty (Labastida-Martinez & Fernandez, 2020). Based on the study of Martín (2007), the perceived low economic benefits from timber might be preventing local farmers from planting native trees and maintaining their agricultural monoculture practices. The policy limitations placed by the government (Executive Order No. 23 Declaring a Moratorium on the Cutting and Harvesting of Timber in the Natural and Residual Forests and Creating the Anti-Illegal Logging Task Force, 2011) on the harvest of native species, even in established woodlots, have had negative impacts on the willingness of community forest groups to plant native trees (Labastida-Martinez & Fernandez, 2020) including the taxation policies that classify cultivated native tree products from natural forests substantially reduced the value of native tree farming systems and cultivation of smallholder native tree farms (Santos Martin & van Noordwijk, 2011).

Local communities living near forest vicinities have been practicing the collection of seeds and wildlings for tree planting as a source of income and livelihood. However, oftentimes, silvicultural practices are not considered in the selection of mother trees or seedling production in nurseries. In the NGP nurseries in Eastern Visayas and Northern Mindanao surveyed by Gregorio et al. (2016), there was widespread production of low-quality planting materials of native tree species using wildlings. Malabrigo, Jr. (2016) emphasized that it is not the absence of planting material and slow growth of native trees that hindered the government to plant native trees but people's lack of awareness. Similarly, poor species-site matching coupled with the presence of exotics in the vicinity and impacts of climate change would affect growth and survival of out planted native tree species in reforestation sites.

Opportunities

According to Le et al. (2014), the quality of seeds or seedlings used in reforestation establishes success as well as the growth quality of trees that survived. The DENR encouraged the communities to establish and develop nurseries through the Barangay Forest Program with the NGP

implementation. It was adopted in Dinagat Islands for seedling production of 10,000 native species (plus 20% mortality allowance) with a funding scheme by the Provincial Environment and Natural Resources Office and the second release of 50% will require the accomplishment of targets (Vista et al., 2016).

The involvement of the academe in reforestation projects of the government proves their competence in research and extension. Developing a scientific way of producing high-quality planting materials of native forest tree species in a short period of time and making it available during planting season is really challenging. The lack of updated reference materials and few experts to identify species of native trees could be opportunities for research and human resource development in the field of botany and taxonomy. The clonal facilities established in each SUC provide an opportunity for students and faculty to conduct research and extension activities on propagation system and protocol development. There are other external fund sources that SUCs can explore as in the case of the Quirino State University that established a clonal nursery facility as part of the DOST-PCAARRD funded project (DOST-PCAARRD, 2019). The facility was able to establish a hardening area and ramet garden of

Bignai (*A. bunius*), Kamagong (*D. blancoi*), Bani (*Pongamia pinnata*), Bolong-eta (*Diospyros pilosanthera* Blanco), Anabiong (*Trema orientalis*), and Lubeg (*Syzygium lineatum* (DC.) Merr.). There are a lot of exploration opportunities for our Philippine native tree species on standards for survival rate, phenology, silviculture, economic value, and the ecosystem services they provide. The academe could collaborate with other government agencies as well as international institutions to conduct forest studies and mangrove reforestation and other similar activities within the scope of NGP implementation of forest development, rehabilitation, maintenance, and protection.

Threats

Government reforestation projects promoted the use of indigenous and endemic tree species but still, a greater proportion of introduced or exotic species was prioritized which are perceived as economically important for furniture, light construction, and ornaments. According to Gregorio et al. (2016), invasive alien species spread by international trade such as exotics threaten native biodiversity. But the ease of collecting seeds or seedlings of exotic species could have contributed to these since most mother trees of exotics have been thriving elsewhere. With the

abundance of invasive plant species, it can displace most of our important native vegetation because they often lack natural predators, or it could trigger the widespread of disease and pest outbreaks when in monoculture plantation.

Natural phenomena like typhoons in the Philippines limit the growth and survival of outplants considering that the planting season is at the same time when typhoons hit the country. Among the ecosystems that will be most vulnerable are the dry forest types in the country with the occurrence of climate change as indicated by the increasing average temperature observed over time (Lasco et al., 2008). Forest and grass fires have been reported during the NGP implementation which could have been brought by the adverse impacts of climate change including human-induced fires in planting sites. Fire incidences tops the list of calamities that destroyed NGP sites as presented by the Commission on Audit (2019) in the NGP performance audit report.

Several studies have been conducted on the effects of climate change on the native tree species distributions in the country using various climate projections and modelling techniques. The results of the studies could be utilized in mainstreaming climate change in the reforestation and forest

management plan to conserve better the potential and suitable habitats of native trees in the country.

Table 3. SWOT analysis of approaches to sustain native trees for reforestation.

	Government Agencies	Local Communities	Non-Government Organization
Strengths	- Priority to native trees as planting materials in protection and production forests - SUCs clonal nursery production augment seedling requirements - Policies and guidelines are in place for the use of native trees in greening activities	- Local knowledge on native trees distribution, propagation, and use - Adoption of Reforestation Farming (RF) - Integration of trees with agricultural crops in agroforestry system - Beneficiaries of reforestation and other related projects	- Enthusiasm in promoting native trees nationwide - Production of available IEC materials on native trees - Established network of native tree nurseries nationwide - Aids in developing capacity of communities and stakeholders
Weakness	- Single or few native species planted (forest planation) - Low survival rate of planted native species - Poor species-site matching	- Decision on what tree species to plant for immediate benefits - Low quality of wildings/poor silvicultural practices in raising seedlings	- Lack reference materials or experts in species identification - Project objectives highly driven by funding resources

	• Allow and accept use of introduced or exotic species • Limited funds to support established SUCs nurseries	• o Use of exotics that provide immediate returns	• o Short-term commitment of members/partners (MOA-based)
• Opportunities	• RDE on vegetative clonal propagation • Funding from external agencies to sustain SUCs clonal operation • Collaboration with international institutions for technical assistance	• Build livelihood or source of income • Partnership with other sectors on greening efforts • o Fund support with the formation of people's organizations (POs)	• Funding from international agencies • Networking and linkages toward integration of conservation efforts • Venue for training of non-experts on native tree species
• Threats	• Seeds and wildlings from mature stands of exotics and introduced species • o Natural phenomena that limit growth and survival of outplants	• Pests and disease occurrence • Laws on tree cutting • Climate change impacts, desertification • Land tenure • o Poverty	• Laws restricting collection of native tree species from forests • o Organizational sustainability

The Framework for Assessment on the Sustainability of Efforts on Native Trees

The framework for assessment on the sustainability of efforts on native trees

in the greening initiative of the Philippines (Figure 4) stems from the NGP implementation of the government which aims to pursue sustainable development for poverty reduction, resource conservation and

protection, climate change mitigation and adaptation by planting some 1.5 billion trees in 1.5 million hectares (2011-2016) as per EO 26, s.2011. In addition to the government objectives with the NGP expansion, it intends to cover all the 7.1 million hectares of remaining unproductive, denuded, and degraded forestlands from 2016 to 2028 as per EO 193, s. 2015. The past greening initiatives of the government such as the Community-Based Forest Management (CBFM), Community-Based Resource Management (CBCRM), and National Forestation Program (NFP) and the ongoing expansion of NGP form the basis in identifying the economic, environmental, and socio-cultural parameters for the sustainability of efforts on native trees. The sustainability parameters on native trees were assessed based on strengths, weaknesses, opportunities, and threats. A very strong point is the promotion and use of native tree species in greening initiatives while a drawback on the quality of native tree species planting materials and decision on what species to plant favoring exotics with immediate benefits. The utilization of native trees in greening programs provides opportunities for collaboration and partnerships of various stakeholders in the implementation, monitoring, and evaluation phases but the natural and human-induced threats pose limits to

the survival of planted native tree species in project sites.

The future direction of native trees in any greening initiative of the Philippines should be supported by proper science-based adaptation policies, strategies and measures; establish partnerships with education institutions, CSOs, and international organizations for support; continuous education in the fields of forestry, agriculture, natural resources to younger generations; sustain the technical, financial, and human resources during the implementation, monitoring, and evaluation of greening projects; and the continuity of activities embedding the system after project termination.

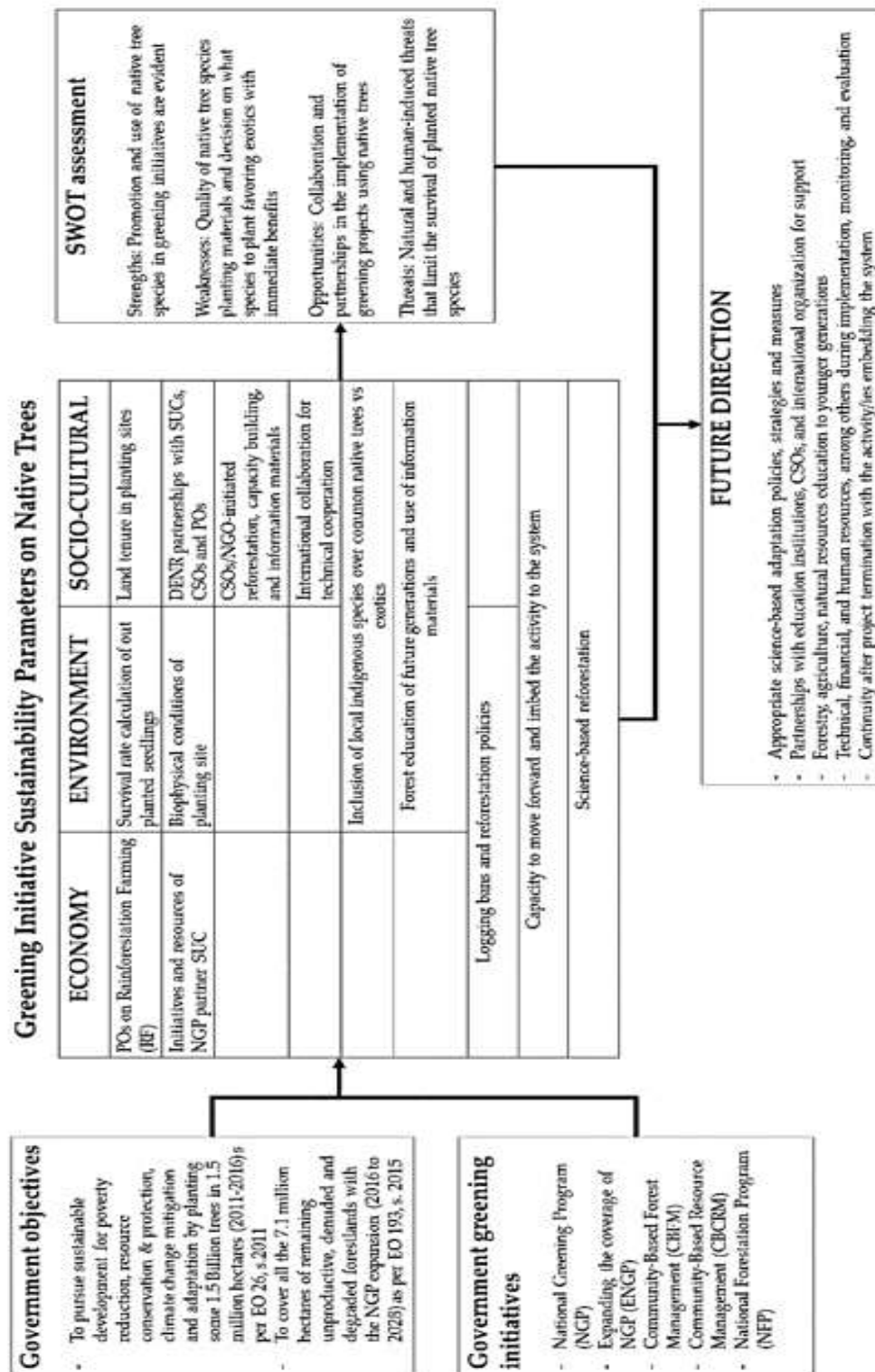


Figure 4. Framework on sustainability of the efforts on native trees

CONCLUSIONS AND RECOMMENDATIONS

One of the Sustainable Development Goals of the 2030 Agenda for Sustainable Development is “to promote the implementation of sustainable management of all types of forests...and substantially increase afforestation and reforestation globally” (United Nations-Department of Economic and Social Affairs, 2020). The use of native tree species for reforestation in the country is just one of the many ways to recover our declining natural resources. DAO 2017-11 has updated the list of Philippine threatened plants and most of our native trees fall either as endangered or vulnerable because of the species’ known use and excessive exploitation without replacement may have contributed to this.

A comprehensive assessment was studied by Le et al. (2012) that covers the initial establishment to forest maturation or the long-term building phase as well as the

physical and non-physical objectives with the potential indicators and drivers of reforestation success. The most common indicators used were 1) the survival rate of planted trees and the area planted compared to a target area for measuring establishment success; 2) tree growth, stand density, stem form of timber trees, and the production of non-timber forest products for measuring forest growth success; 3) vegetation structure, species diversity, and ecosystem functions for measuring environmental success; and 4) local income and employment opportunities, other livelihood sources, supply of food and fiber, stability of market prices of local products, and local empowerment and capacity building for measuring socio-economic success (Le et al. 2012). For the potential drivers, Le et al. (2012) identified the factors that relate to technical and biophysical, socio-economic, institutional, policy, and management; and the

characteristics of the reforestation project.

The government has been consistent in implementing various programs to rehabilitate our denuded forests and the use of native tree species for reforestation is prioritized over introduced or exotic species. The academe's research and development on these species could further promote their use by immediate beneficiaries. Establishing partnerships with civil society organizations and local communities has contributed to the achievement of set objectives. So, crafting greening strategies should not only focus on survival rates and completion of targets but consider the diversification of methods employed to ensure the efficiency of use and sustainability of our native tree species for ecosystem diversity. Also, the success of conserving native trees does not only rely on mass propagation nor quality planting material production, but it is also important to recognize the role of

academe, local communities including non-government efforts as integral in all greening initiatives led by the government. The cooperation between these key sectors is essential in achieving reforestation success that aim to promote utilization of native species for the greater purpose of conserving and protecting our rich biodiversity to mitigate climate change impacts, enhance productivity, and alleviate poverty.

For successful reforestation in Philippine forestry, changing the paradigm from a plantation approach to Rainforestation Farming (RF) makes sound and sustainable biodiversity conservation as well as rural development. Ground-based, inter-linked efforts of forest-dependent communities, local agencies, and committed partner agencies and institutional support contribute to reforestation success which as evident in several regions of Southeast Asia (Milan, 2020) and the RF strategy has been proven to enhance biodiversity,

soil health, watershed function, and wildlife conservation.

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